

EE5346 – 自主机器人导航 (2026 年春季)

作业 #5

MLE 与扩展卡尔曼滤波器

问题 1: 最大似然估计 (MLE)

测量方程为

$$z_k = x + n_k, \quad n_k \sim \mathcal{N}(0, \sigma^2), \quad k = 1, 2, \dots, N,$$

其中 x 为待估计的机器人位置 (未知常数), n_k 为零均值高斯噪声 (题目中的 $\mathcal{N}(0, \sigma)$ 中 σ 视为标准差, 方差为 σ^2), N 个测量相互独立。由 $z_k | x \sim \mathcal{N}(x, \sigma^2)$, 单测量的概率密度函数为

$$p(z_k | x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(z_k - x)^2}{2\sigma^2}\right].$$

似然函数 $L(x) = p(Z | x)$

由各测量独立, 联合 PDF 等于各 PDF 的乘积:

$$L(x) = \prod_{k=1}^N p(z_k | x) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left[-\frac{1}{2\sigma^2} \sum_{k=1}^N (z_k - x)^2\right].$$

即

$$L(x) = (2\pi\sigma^2)^{-N/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{k=1}^N (z_k - x)^2\right].$$

负对数似然 $\mathcal{L}(x) = -\ln L(x)$

利用 $\ln(ab) = \ln a + \ln b$ 与 $\ln(e^c) = c$:

$$\mathcal{L}(x) = \frac{N}{2} \ln(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{k=1}^N (z_k - x)^2.$$

第一项与 x 无关，因此

$$x^* = \arg \min_x \mathcal{L}(x) \equiv \arg \min_x \sum_{k=1}^N (z_k - x)^2,$$

即在高斯噪声下，MLE 等价于最小二乘问题。

求解 MLE x^*

对 $\mathcal{L}(x)$ 求导并令其为零：

$$\frac{d\mathcal{L}}{dx} = \frac{1}{\sigma^2} \sum_{k=1}^N (x - z_k) = 0 \implies Nx - \sum_{k=1}^N z_k = 0.$$

二阶导数 $\frac{d^2\mathcal{L}}{dx^2} = \frac{N}{\sigma^2} > 0$ ，确认为极小值。因此

$$x^* = \frac{1}{N} \sum_{k=1}^N z_k.$$

即在 i.i.d. 高斯噪声下，机器人位置的 MLE 等于所有测量值的样本均值。

数值结果

代入 $N = 3$, $z_1 = 3$, $z_2 = 4$, $z_3 = 7$ ：

$$x^* = \frac{3 + 4 + 7}{3} = \frac{14}{3} \approx 4.667.$$

问题 2：教材式 (8.32) 的推导

考虑非线性运动模型（加性过程噪声形式）

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k),$$

其中 $\mathbf{w}_k$ 与历史状态独立。上一时刻的后验为

$$\mathbf{x}_{k-1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k-1}, \hat{\mathbf{P}}_{k-1}).$$

在估计点处一阶泰勒展开

由于 f 非线性，EKF 在 $\hat{\mathbf{x}}_{k-1}$ 处线性化：

$$f(\mathbf{x}_{k-1}, \mathbf{u}_k) \approx f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) + \mathbf{F}(\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1}), \quad \mathbf{F} = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k}.$$

代入状态方程：

$$\mathbf{x}_k \approx f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) + \mathbf{F}(\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1}) + \mathbf{w}_k. \quad (\star)$$

预测均值 $\tilde{\mathbf{x}}_k$

由均值（向量函数期望）的定义 $\tilde{\mathbf{x}}_k = E[\mathbf{x}_k]$ ，对 $(\star)$ 两边取期望并利用线性性：

$$\tilde{\mathbf{x}}_k = f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) + \mathbf{F} \underbrace{E[\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1}]}_{=0} + \underbrace{E[\mathbf{w}_k]}_{=0}.$$

故

$$\boxed{\tilde{\mathbf{x}}_k = f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k).}$$

预测协方差 $\tilde{\mathbf{P}}_k$

记误差 $\delta \mathbf{x}_{k-1} = \mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1}$ 。由 $(\star)$ 及上一节结果：

$$\mathbf{x}_k - \tilde{\mathbf{x}}_k = \mathbf{F} \delta \mathbf{x}_{k-1} + \mathbf{w}_k.$$

按协方差矩阵的定义 $\tilde{\mathbf{P}}_k = E[(\mathbf{x}_k - \tilde{\mathbf{x}}_k)(\mathbf{x}_k - \tilde{\mathbf{x}}_k)^T]$ 展开：

$$\tilde{\mathbf{P}}_k = E[\mathbf{F} \delta \mathbf{x}_{k-1} \delta \mathbf{x}_{k-1}^T \mathbf{F}^T + \mathbf{F} \delta \mathbf{x}_{k-1} \mathbf{w}_k^T + \mathbf{w}_k \delta \mathbf{x}_{k-1}^T \mathbf{F}^T + \mathbf{w}_k \mathbf{w}_k^T].$$

由于 $\mathbf{w}_k$ 与 $\mathbf{x}_{k-1}$ 独立， $E[\delta \mathbf{x}_{k-1} \mathbf{w}_k^T] = \mathbf{0}$ ，交叉项消失；又

$$E[\delta \mathbf{x}_{k-1} \delta \mathbf{x}_{k-1}^T] = \hat{\mathbf{P}}_{k-1}, \quad E[\mathbf{w}_k \mathbf{w}_k^T] = \mathbf{R}_k,$$

因此

$$\boxed{\tilde{\mathbf{P}}_k = \mathbf{F} \hat{\mathbf{P}}_{k-1} \mathbf{F}^T + \mathbf{R}_k.}$$

关于非加性噪声的说明

若噪声以 $f(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k)$ 方式进入模型（如本作业第 3 题中过程噪声乘以输入耦合矩阵），则需引入噪声雅可比

$$\mathbf{G} = \left. \frac{\partial f}{\partial \mathbf{w}} \right|_{\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k, \mathbf{0}},$$

此时预测协方差为

$$\tilde{\mathbf{P}}_k = \mathbf{F} \hat{\mathbf{P}}_{k-1} \mathbf{F}^T + \mathbf{G} \mathbf{Q} \mathbf{G}^T.$$

教材式 (8.32) 通过将 $\mathbf{G} \mathbf{Q} \mathbf{G}^T$ 整体记为 $\mathbf{R}_k$ ，与上述加性形式一致。

问题 3: EKF 雅可比矩阵 $\mathbf{F}$ 与 $\mathbf{H}$

记状态、输入、地标分别为

$$\mathbf{x}_k = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix}, \quad \mathbf{u}_k = \begin{bmatrix} v_k \\ \omega_k \end{bmatrix}, \quad \mathbf{y}^l = \begin{bmatrix} x_l \\ y_l \end{bmatrix}.$$

运动模型 $\Rightarrow$ 雅可比 $\mathbf{F}$

题目给定的运动模型为

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \Delta t \begin{bmatrix} \cos \theta_{k-1} & 0 \\ \sin \theta_{k-1} & 0 \\ 0 & 1 \end{bmatrix} (\mathbf{u}_k + \mathbf{w}_k), \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}).$$

取 $\mathbf{w}_k = \mathbf{0}$ 用于求 $\mathbf{F}$, 逐分量展开:

$$x_k = x_{k-1} + \Delta t v_k \cos \theta_{k-1},$$

$$y_k = y_{k-1} + \Delta t v_k \sin \theta_{k-1},$$

$$\theta_k = \theta_{k-1} + \Delta t \omega_k.$$

一阶近似形式

在 $(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k)$ 处展开:

$$f(\mathbf{x}_{k-1}, \mathbf{u}_k) \approx f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) + \mathbf{F} (\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1}).$$

求 $\mathbf{F} = \partial f / \partial \mathbf{x}_{k-1}$

逐项求偏导:

$$\begin{aligned} \frac{\partial x_k}{\partial x_{k-1}} &= 1, & \frac{\partial x_k}{\partial y_{k-1}} &= 0, & \frac{\partial x_k}{\partial \theta_{k-1}} &= -\Delta t v_k \sin \theta_{k-1}, \\ \frac{\partial y_k}{\partial x_{k-1}} &= 0, & \frac{\partial y_k}{\partial y_{k-1}} &= 1, & \frac{\partial y_k}{\partial \theta_{k-1}} &= \Delta t v_k \cos \theta_{k-1}, \\ \frac{\partial \theta_k}{\partial x_{k-1}} &= 0, & \frac{\partial \theta_k}{\partial y_{k-1}} &= 0, & \frac{\partial \theta_k}{\partial \theta_{k-1}} &= 1. \end{aligned}$$

因此

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & -\Delta t v_k \sin \hat{\theta}_{k-1} \\ 0 & 1 & \Delta t v_k \cos \hat{\theta}_{k-1} \\ 0 & 0 & 1 \end{bmatrix}.$$

矩阵在估计点 $\hat{\theta}_{k-1}$ 与里程计输入 v_k 处求值。

测量模型 $\Rightarrow$ 雅可比 $\mathbf{H}$

中间变量

定义

$$\begin{aligned}\Delta x &= x_l - x_k - d \cos \theta_k, & \Delta y &= y_l - y_k - d \sin \theta_k, \\ r &= \sqrt{\Delta x^2 + \Delta y^2}, & r^2 &= \Delta x^2 + \Delta y^2.\end{aligned}$$

测量模型为

$$h(\mathbf{x}_k, \mathbf{y}^l) = \begin{bmatrix} r \\ \text{atan2}(\Delta y, \Delta x) - \theta_k \end{bmatrix}.$$

一阶近似形式

在预测位姿 $\tilde{\mathbf{x}}_k$ 处展开：

$$h(\mathbf{x}_k, \mathbf{y}^l) \approx h(\tilde{\mathbf{x}}_k, \mathbf{y}^l) + \mathbf{H}(\mathbf{x}_k - \tilde{\mathbf{x}}_k).$$

辅助偏导

$$\begin{aligned}\frac{\partial \Delta x}{\partial x_k} &= -1, & \frac{\partial \Delta y}{\partial y_k} &= -1, \\ \frac{\partial \Delta x}{\partial \theta_k} &= d \sin \theta_k, & \frac{\partial \Delta y}{\partial \theta_k} &= -d \cos \theta_k.\end{aligned}$$

(a) 距离 r 部分

$$\begin{aligned}\frac{\partial r}{\partial x_k} &= \frac{1}{r}(\Delta x \cdot (-1) + \Delta y \cdot 0) = -\frac{\Delta x}{r}, & \frac{\partial r}{\partial y_k} &= -\frac{\Delta y}{r}, \\ \frac{\partial r}{\partial \theta_k} &= \frac{1}{r}(\Delta x \cdot d \sin \theta_k + \Delta y \cdot (-d \cos \theta_k)) = \frac{d}{r}(\Delta x \sin \theta_k - \Delta y \cos \theta_k).\end{aligned}$$

(b) 方位 $\phi = \text{atan2}(\Delta y, \Delta x) - \theta_k$ 部分

利用

$$\frac{\partial \text{atan2}(v, u)}{\partial u} = -\frac{v}{u^2 + v^2}, \quad \frac{\partial \text{atan2}(v, u)}{\partial v} = \frac{u}{u^2 + v^2},$$

通过链式法则得：

$$\begin{aligned} \frac{\partial \phi}{\partial x_k} &= -\frac{\Delta y}{r^2} \cdot (-1) + \frac{\Delta x}{r^2} \cdot 0 = \frac{\Delta y}{r^2}, \\ \frac{\partial \phi}{\partial y_k} &= -\frac{\Delta y}{r^2} \cdot 0 + \frac{\Delta x}{r^2} \cdot (-1) = -\frac{\Delta x}{r^2}, \\ \frac{\partial \phi}{\partial \theta_k} &= -\frac{\Delta y}{r^2} \cdot d \sin \theta_k + \frac{\Delta x}{r^2} \cdot (-d \cos \theta_k) - 1 \\ &= -\frac{d}{r^2}(\Delta x \cos \theta_k + \Delta y \sin \theta_k) - 1. \end{aligned}$$

最终 $\mathbf{H}$ (2×3 矩阵)

$$\mathbf{H} = \begin{bmatrix} -\frac{\Delta x}{r} & -\frac{\Delta y}{r} & \frac{d}{r}(\Delta x \sin \theta_k - \Delta y \cos \theta_k) \\ \frac{\Delta y}{r^2} & -\frac{\Delta x}{r^2} & -\frac{d}{r^2}(\Delta x \cos \theta_k + \Delta y \sin \theta_k) - 1 \end{bmatrix},$$

其中

$$\Delta x = x_l - x_k - d \cos \theta_k, \quad \Delta y = y_l - y_k - d \sin \theta_k, \quad r = \sqrt{\Delta x^2 + \Delta y^2}.$$

所有项在预测位姿 $\tilde{\mathbf{x}}_k = (\tilde{x}_k, \tilde{y}_k, \tilde{\theta}_k)$ 与已知地标 $\mathbf{y}^l$ 处求值。

答案速览

题号 关键结果

$$1 \quad L(x) = (2\pi\sigma^2)^{-N/2} \exp\left[-\frac{1}{2\sigma^2} \sum_k (z_k - x)^2\right]$$

$$1 \quad \mathcal{L}(x) = \frac{N}{2} \ln(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_k (z_k - x)^2$$

$$1 \quad x^* = \frac{1}{N} \sum_k z_k = \frac{14}{3} \approx 4.667$$

$$2 \quad \tilde{\mathbf{x}}_k = f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k)$$

$$2 \quad \tilde{\mathbf{P}}_k = \mathbf{F}\hat{\mathbf{P}}_{k-1}\mathbf{F}^T + \mathbf{R}_k$$

$$3 \quad \mathbf{F} = \begin{bmatrix} 1 & 0 & -\Delta t v_k \sin \hat{\theta}_{k-1} \\ 0 & 1 & \Delta t v_k \cos \hat{\theta}_{k-1} \\ 0 & 0 & 1 \end{bmatrix}$$

$$3 \quad \mathbf{H} = \begin{bmatrix} -\Delta x/r & -\Delta y/r & (d/r)(\Delta x \sin \theta_k - \Delta y \cos \theta_k) \\ \Delta y/r^2 & -\Delta x/r^2 & -(d/r^2)(\Delta x \cos \theta_k + \Delta y \sin \theta_k) - 1 \end{bmatrix}$$
